

Information Disclosure and Environmental Improvement

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Learning Objectives

- After taking this class, you will be able to：
  - understand the importance of information in environmental governance
  - understand the practices dealing with information asymmetry in environmental governance
  - understand the mechanisms of information disclosure on environmental improvement in the Chinese context

Why is information important?

Information in Economic Activities

- Market Failure
  - an inefficient distribution of goods and services in the free market
  - Typical causes of market failure
    - Externality
    - Public goods
    - Monopoly
    - Information asymmetries

- Full and complete information is a precondition of the efficacy of markets.
- Information asymmetry would generate issues of adverse selection and moral hazard.

The Importance of Information

- **A Warm-up Activity**
- Extra credit question

Select whether you want 1 point, or 2 points added onto your final exam grade. But if more than 10% of the class selects 2 points, then no one gets any points.

*Originally posted by a professor from the University of Maryland

- What is your decision?
- Suppose you have an opportunity to communicate with each other about your decision. Does it change your choice?
- What will you do if you can observe each other's decision after the communication?

The Importance of Information

- **Information Asymmetry and Its Consequences**

- **Information asymmetry** is an *imbalance* between two negotiating parties in their knowledge of relevant factors and details.

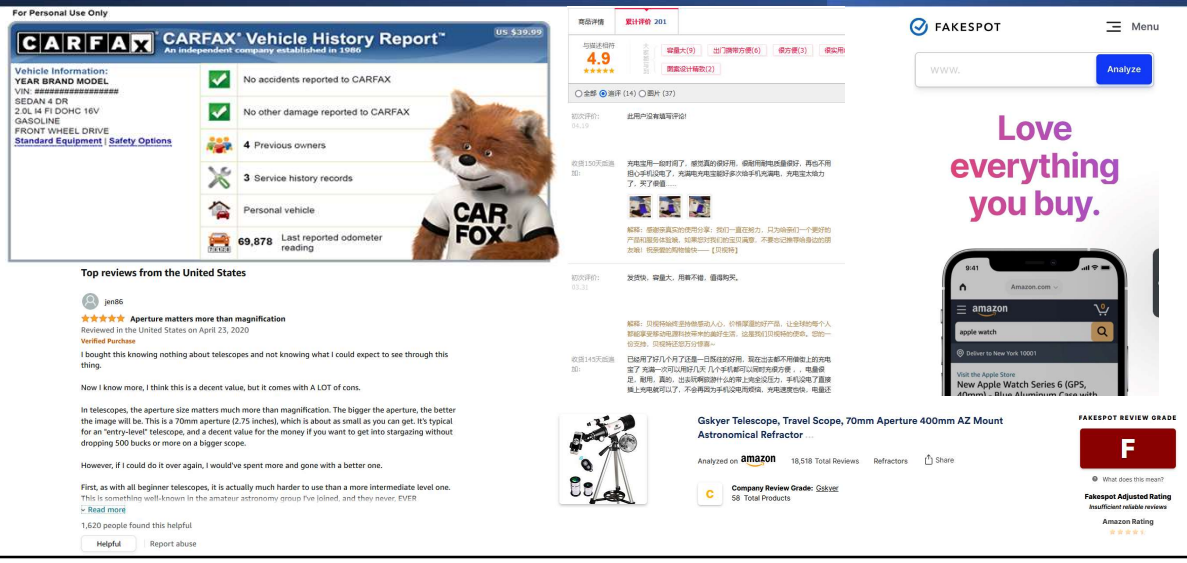


Adverse Selection



With information asymmetry, we cannot identify the enterprises with better environmental performance.

The Importance of Information



The Importance of Information

- **Information Asymmetry and Its Consequences**

- **Moral hazard** is a term describing how behavior changes when people are insured against losses.
- Principal-agent problem, also called as agency problem
- *We talked a lot about policy interests and preferences of different actors. PA problems emerge due to the conflict of interests.*



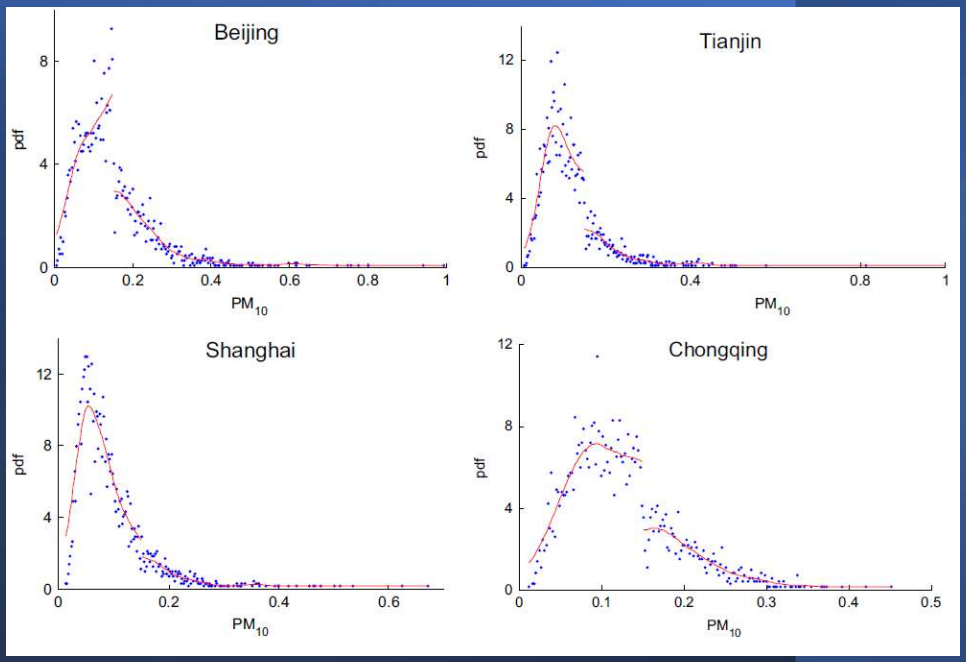
Moral Hazard



With information asymmetry, higher-level governments cannot effectively figure out whether their agents (local governments) cheat on their environmental policy implementation.

The Importance of Information

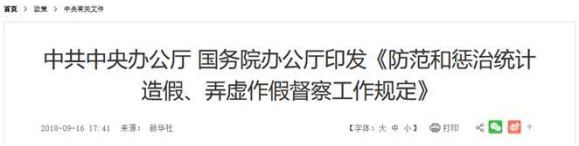
- **Information Asymmetry in Environmental Governance: Gaming the Pollution Data**
  - 113 key cities have been ranked in the Annual Quantified Assessment of Urban Environmental Improvement (*Chengkao*) since 1989.
  - The rankings of environmental performance by the *Chengkao* were intended to award the title of "Environmental Protection Model City" to top performing cities.
  - Air quality is the single most important indicator in the assessment, which accounts for 20 percent of a city's environmental quality grade.
  - Using officially reported API data from 37 large cities during 2000-2009, Chen et al (2012) find a significant discontinuity at the threshold of 100.
  - The model cities were less likely to report API right above 100 when they were close to the targeted blue-sky days in the fourth quarter of the year when or before they won the model city award.
  - In another research, scholars examine the same data manipulation issue on 113 cities over the period 2001-2010. They also find sharp discontinuities at the cut-off for blue-sky days for PM10 data for 50% of the cities. (Ghanem and Zhang, 2014)



The Importance of Information



“Zero tolerance” to statistical manipulation



“Regulations on the Prevention and Punishment of Statistical Fraud and Falsification Inspection Work” issued by The General Office of the Central Committee of the Communist Party of China and The General Office of the State Council

How to deal with information asymmetry and moral hazard?

Practices Dealing with Information Asymmetry

- **Intensify Central Monitoring**
- National Specially Monitored Enterprises
  - Since 2007, the MEP and EPBs of provinces annually publish lists that identify the most polluting firms.
  - The initial NSM firms were selected on the basis of their pollutant emissions over the last two years.
  - All polluting firms with records in the CES were ranked by their COD and NH3-N emissions in 2005. The top-ranked firms, which accounted for 65% of total emissions of either COD or NH3-N, were designated as NSM water-polluting firms.
  - The information about pollution discharge of these enterprises is directly uploaded to MEP.
- Real-time monitoring system

Number of State-monitored Enterprises by Type and by Year

Year	sewage	Sewage treatment plant	air	Heavy metal	Animal farm	Dangerous Waste
2007	3115	658	3592	---	---	---
2008	3115	658	3592	---	---	---
2009	3945	1296	3715	---	---	---
2010	4547	1814	3472	---	---	---
2011	4852	2872	4452	---	---	---
2012	5159	2883	3605	3732	---	---
2013	4944	3581	4189	2834	249	---
2014	4001	3606	3865	2771	167	---
2015	2937	3788	3268	3432	52	1443
2016	2660	3812	3281	2901	21	1637
2017	2504	3991	3365	2535	20	1785

Practices Dealing with Information Asymmetry

- **Intensify Central Monitoring**

Comparing NSM Enterprises and Non-NSM Enterprises

Pollutants	Year	Non-National Specially Monitored Enterprises		National Specially Monitored Enterprises	
		Observation	Mean	Observation	Mean
COD	2005	18,964	29.04	1643	1203
	2006	18,964	35.54	1643	1127
	2007	18,964	36.74	1643	1020
	2008	18,964	34.10	1643	878.1
	2009	18,964	33.17	1643	790.4
NH3-N	2005	18,964	1.8	1643	133.3
	2006	13,282	3.29	1316	120.1
	2007	13,098	3.26	1301	92.7
	2008	13,857	2.85	1357	70.86
	2009	15,253	2.53	1475	53.32

(Zhang et al, 2018)

Practices Dealing with Information Asymmetry

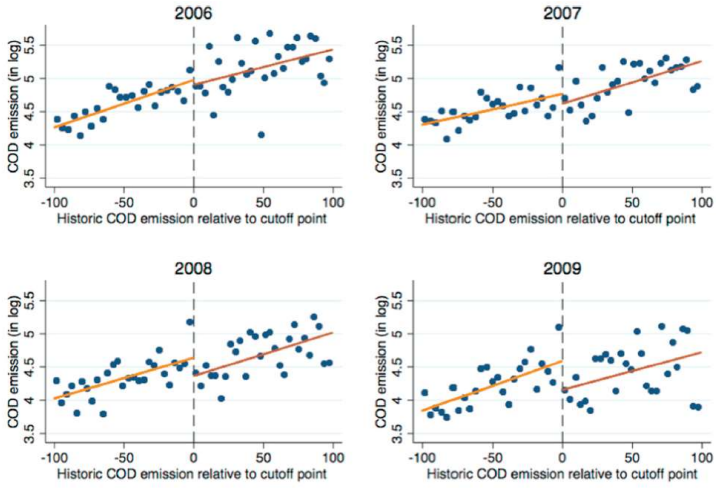
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Practices Dealing with Information Asymmetry

- Does the program of national specially monitored enterprises work? Yes.

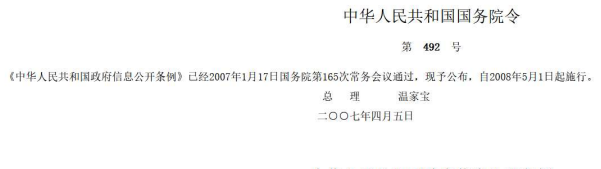


Regression Discontinuity Design

(Zhang et al, 2018)

Practices Dealing with Information Asymmetry

- **Open Government Information Regulation and PITI**



中华人民共和国国务院令

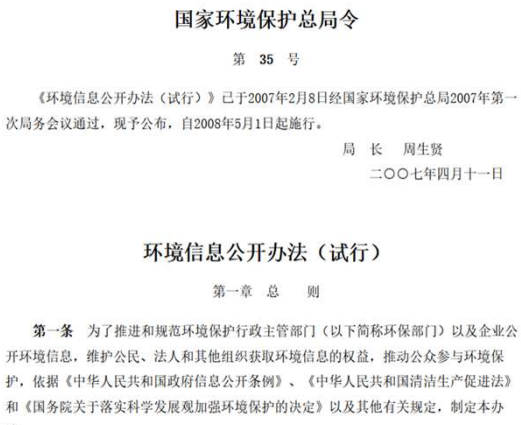
第 492 号

《中华人民共和国政府信息公开条例》已经2007年1月17日国务院第165次常务会议通过，现予公布，自2008年5月1日起施行。

总 理 温家宝

二〇〇七年四月五日

中华人民共和国政府信息公开条例



国家环境保护总局令

第 35 号

《环境信息公开办法（试行）》已于2007年2月8日经国家环境保护总局2007年第一次局务会议通过，现予公布，自2008年5月1日起施行。

局 长 周生贤

二〇〇七年四月十一日

环境信息公开办法（试行）

第一章 总 则

第一条 为了推进和规范环境保护行政主管部门（以下简称环保部门）以及企业公开环境信息，维护公民、法人和其他组织获取环境信息的权益，推动公众参与环境保护，依据《中华人民共和国政府信息公开条例》、《中华人民共和国清洁生产促进法》和《国务院关于落实科学发展观加强环境保护的决定》以及其他有关规定，制定本办法。

- Open Government Information Regulations (OGIR) is initiated in 2008.
- MEP was one of the earliest governmental department that implement OGIR.
- These rules require all local governments to disclose information to the public.

Practices Dealing with Information Asymmetry

- **Open Government Information Regulation and PITI**

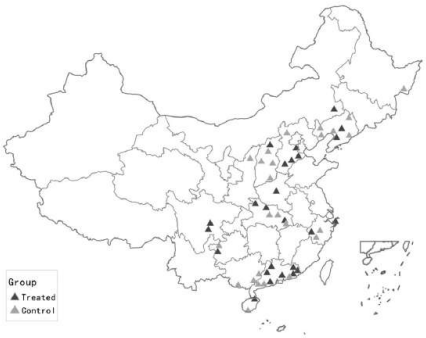
- IPE and Pollution Information Transparency Index (PITI)
  - The first PITI index was released in June 2009, assessing performance one year after the OGIR took effect.
  - Two NGOs, IPE and the Natural Resources Defense Council (NRDC) began conducting annual evaluations of environmental transparency in 113 Chinese cities.
  - Each city received a score between 0 and 100 on this Pollution Information Transparency Index (PITI).
  - 60 points were assigned for behaviors mandated by law, with additional points assigned for other steps improving public access to information.

序号	省区	城市	历年均分	2008	2009-2010	2011	2012	2013-2014	2014-2015	2015-2016	2016-2017	2017-2018	2018-2019
1	浙江	宁波	74.3	72.9	82.1	83.7	85.3	65.9	69.1	72.8	70	77.7	69.0
2	北京	北京	68.0	49.1	43.5	72.9	72.9	58.7	67	77.1	75.7	79.6	76.0
3	浙江	温州	67.8	53.3	56.5	72.7	70.4	53.2	69.3	72.7	78.1	81	69.2
4	山东	青岛	66.3	38.4	37.7	70.6	74.4	55.8	66.8	74.8	75.1	78.5	78.8
5	浙江	杭州	65.0	48	36.8	60.2	70.8	53.1	65.2	75.9	72.5	75	78.8
6	上海	上海	65.0	56.5	67.2	68.8	65.6	53	64.6	66.2	71.6	66.4	67.5
7	浙江	台州	63.8	48.4	66.6	75.4	58.1	43.4	59.4	67.7	66.5	74.3	71.2
8	广东	深圳	63.7	51.1	74.5	83.3	73.1	35.4	47.6	64	69	65	68.8
9	江苏	常州	63.5	56.8	65.8	76.8	60.3	47.6	52.5	59.8	63	70.3	72.7
10	广东	广州	62.4	44.4	51.9	61.2	71.4	34	46	71.9	76.9	67.5	80.8
11	福建	福州	61.6	63.7	62.5	68.1	67.4	41.8	55.4	51	57.6	66.3	71.9
12	广东	中山	61.2	42.9	66.4	76	63.8	33.8	40.4	67.4	73.4	71.6	68.5
13	广东	东莞	61.1	34.3	58.5	72.1	74.9	30.3	52.6	55.6	70.2	76.3	73.8
14	安徽	合肥	60.7	66.6	56.8	55.2	57.1	38.9	60.1	53.2	67.3	63.7	74.3
15	广东	佛山	60.4	44.4	70.3	74.6	53.5	26.4	48.5	59.3	69.8	68.4	74.4
16	山东	烟台	60.3	44.5	48.7	37.3	51.3	42.8	68.9	65.4	66	75	81.4
17	江苏	苏州	60.1	47	60.3	60.1	63.8	42.5	60.3	67.8	72.2	59.5	63.7
18	江苏	南京	59.4	47.2	58.4	62.7	65.5	50.9	66.5	55.7	63.6	64.5	59.1
19	江苏	南通	58.7	56.2	61.9	63.1	63.8	36.8	62.3	58.3	61.6	61.9	60.1
20	江苏	无锡	58.4	51.6	54.3	60.3	57.7	42.1	54.3	54.5	59.3	72.8	68.0

Practices Dealing with Information Asymmetry

- **Open Government Information Regulation and PITI**
- Whether NGO monitoring increases transparency?
  - Draw a purposeful sample of cities not previously monitored, but which are expected that PITI would have the greatest potential.
  - From the cities not included in prior PITI monitoring and releases, we selected the 50 cities with the highest predicted PITI scores based on out-of-sample predictions.
  - Randomly assigned 25 of the 50 sampled municipalities to a public disclosure of their PITI score, forming matched-pair blocks based on an ordering of the pretreatment PITI scores for assignment.

FIGURE 1 Assignment of Treatment and Control among Experimental Sample of Cities

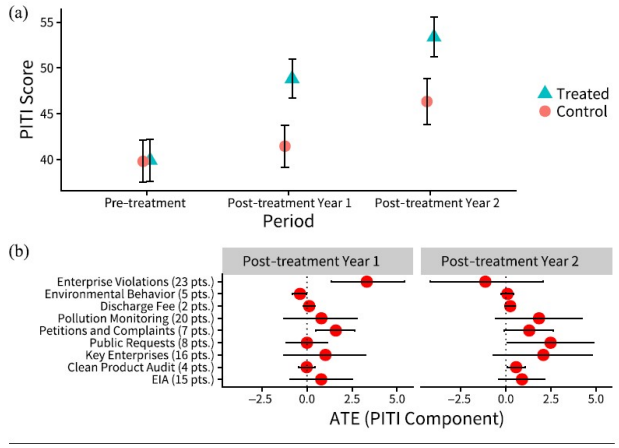


(Anderson, Buntaine, Liu, and Zhang, 2019)

Practices Dealing with Information Asymmetry

- **Open Government Information Regulation and PITI**
- NGO monitoring and Transparency: Research Design
  - Replicated the PITI scoring process using data from 2014 and 2015/2016 (two post-treatment years), with cross-checking from IPE prior to public dissemination.
  - The treatment is the disclosure of the PITI score by IPE and the reinforced dissemination by the research team.
  - The PITI scores for cities in the treatment group were published in a variety of ways, including as part of IPE's annual PITI release, online (via WeChat and Weibo), and direct mailings of reports to local environmental protection bureaus assigned to the treatment.
  - The PITI scores for cities in the control group were not released publicly or revealed to the cities.

Practices Dealing with Information Asymmetry

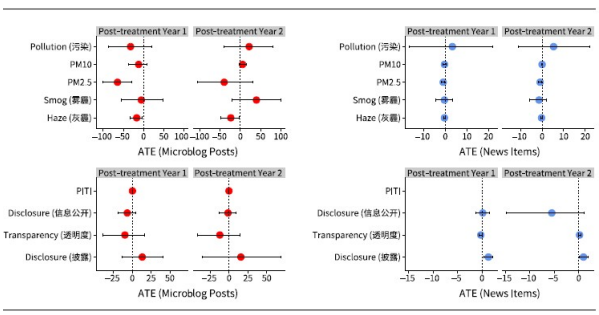


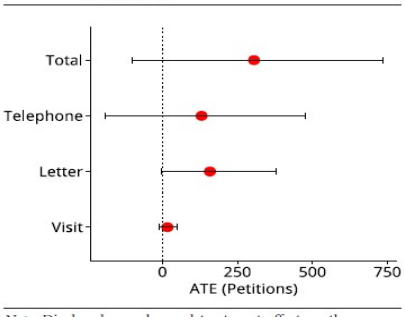
(Anderson, Buntaine, Liu, and Zhang, 2019)

- The treatment, public disclosure of PITI scores, increases compliance of city governments with mandates to be transparent about the management of pollution.
- On average, cities in the treatment group have transparency scores that are approximately 7 points higher on a scale of 0-100 than the control cities.
- Treated cities increase the disclosure of information about enterprises violations of pollution standards as compared to the control cities.
- Treated cities increase transparency about the disposition of citizen petitions and complaints.

Practices Dealing with Information Asymmetry

- The Mechanisms: citizen's attention to environmental issues





- There is little evidence that NGO disclosure of information about compliance increases public discontent.
- There is little evidence that PITI facilitates the direct contact of citizens with municipal governments.

(Anderson, Buntaine, Liu, and Zhang, 2019)

Practices Dealing with Information Asymmetry

- The Mechanisms: Reduced monitoring costs
  - 20 interviews with local EPB officials and 7 local NGOs
  - Local officials in both experimental conditions are clearly sensitive to the directives coming from higher levels of government about the disclosure of information to the public.
  - Localities are sensitive to being found out of compliance.
  - PITI evaluation did help increase the frequency of communication between local NGOs and local governments.

(Anderson, Buntaine, Liu, and Zhang, 2019)

Practices Dealing with Information Asymmetry

- **Involving Third-party Monitoring**

- Can you recall the number of environmental reporting hotline?



- Do you know any other governmental reporting hotline?

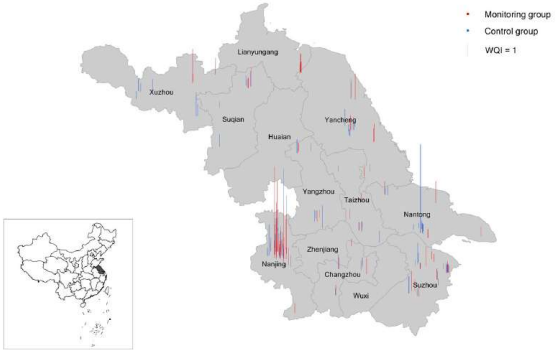


Practices Dealing with Information Asymmetry

- **Involving Third-party Monitoring**

Practices Dealing with Information Asymmetry

- **Involving Third-party Monitoring**



- Randomly assigned half of 160 urban waterways in Jiangsu province to be monitored by a partner non-governmental organization for 15 months
- Treatments:
  - Dissemination of monitoring results to governmental organizations
  - Dissemination to the public living near waterways by put up posters where they would be most noticeable to residents

(Buntaine, Zhang, and Hunnicutt, 2021)

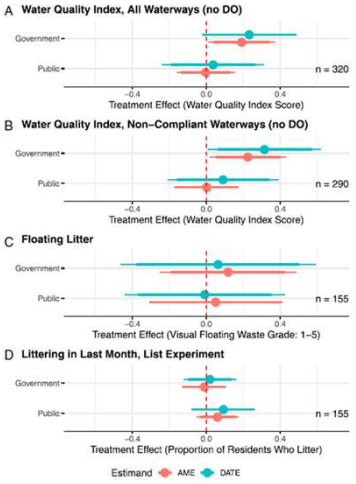
Practices Dealing with Information Asymmetry

- **Involving Third-party Monitoring**

- Water ways assigned to the government dissemination treatment experienced at least a 19% improvement in water quality, on average.

- The positive effect of the government dissemination treatment is most apparent in pollutants with the highest levels of noncompliance at baseline.

- The government dissemination treatment did not have detectable effects on the amount of floating litter or littering behavior by residents.



(Buntaine, Zhang, and Hunnicutt, 2021)

Practices Dealing with Information Asymmetry

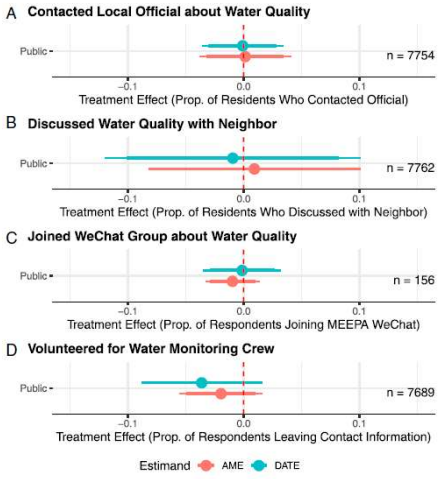
- **Involving Third-party Monitoring**

- The public dissemination treatment did not have a detectable effect on chemical water quality.

- It did not have a detectable effect on the amount of littering.

- It also has no detectable effects on these self-reported and reveal behaviors, which are expected to be the intermediate steps to improved water quality.

- Information is not the primary limit on volunteerism and petitioning related to pollution.



(Buntaine, Zhang, and Hunnicutt, 2021)

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